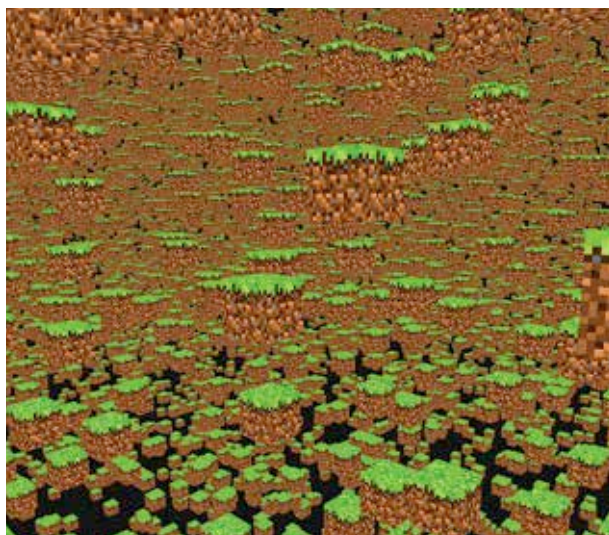


Courtesy: Lucas Camolezi



ordered is a highly complex task, and this problem tormented Ken Perlin. He hated the way CGI back in the 80s looked so structured, especially with natural things like terrain and smoke. So he developed an algorithm that would create random but structured patterns, called Perlin Noise, which became the industry standard for such applications. The way Minecraft uses Perlin noise is by creating a 2D height map using Perlin noise. Each pixel value in the Perlin noise represents a specific height used by the game engine to extrapolate a mesh to that height in 3D. Adding some more basic rules (for example, the game should add snow to areas above a certain height) could create some decent looking terrain for a game. This only resolved the terrain generation, but what about cave systems?

Building cave systems and cliffs with overhangs was another challenge that required a unique approach. To generate a cave, Minecraft utilised something called a Perlin worm. The idea is simple: Pick a random point in 3D and mark that as the point where the cave terminates and navigate through the terrain in a noisy way,

subtracting chunks and creating a cave till you make your way out to the surface. This requires a 3D Perlin noise map, with the 3rd dimension signifying the angle where the Perlin worm should branch off into.

Procgen is not a magic wand that game developers can use to shoo away all their design problems and reduce their development time. Developing procedurally generating systems that are robust and create content that is always engaging is a big challenge by itself. Procedural level creation still requires a designer to fine-tune the limits and parameters of game level building blocks, but a lot of the work that a level designer used to do will now be done by a programmer. The developers often lose sight that having a lot of variety doesn't always equate to being a better experience.

Hello Games learnt it the hard way when they released No Man's Sky in 2016. No Man's Sky was released



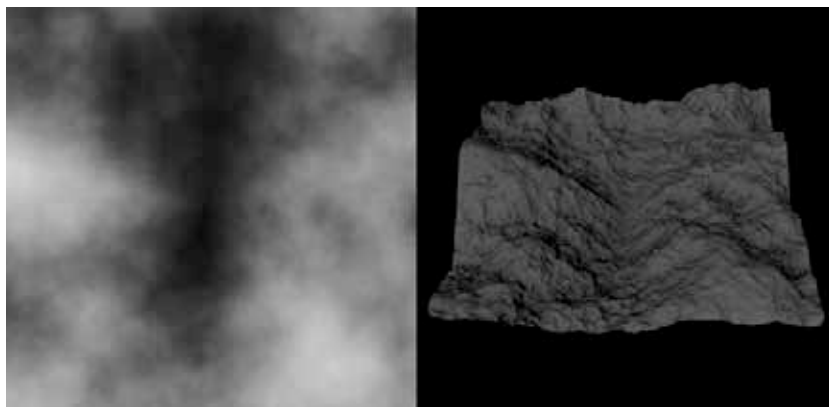
Perlin worm at work

on PlayStation 4 and P.C. after years of anticipation; however it fell short of expectations. The game was criticised for various reasons, including mindless resource-gathering gameplay, planets that all looked the same, and lonely alien life forms that could always be found in the same places, saying the same things. Since then, it has improved significantly, having made a resurgence last year for one of the most improved games since launch.

Roguelike games

Roguelike is a genre of games that builds upon the concept of RNG and ProcGen and heavily relies on it. The two major components that define a roguelike game are perma-death and randomly generated dungeons, adopted from a 1980 game called – Rogue and hence the name “rogue-like”. The term comes from old discussion boards, where people hunting for games similar to Rogue called them roguelikes. A lot more goes into developing a roguelike game than just these two elements, and some developers tried to narrow down all of the elements, which eventually became the Berlin interpretation of roguelikes. They had 8 ‘high-value’ factors and 6 low value ‘factors’. Each element of the Berlin Interpretation plays off of each other well.

That was a lot of terminologies that make very little sense unless you're heading the game design department of Supergiant games, so let's simplify: Roguelikes involves random dungeon generation, mystery loot, and perma-



On Left: Perlin Noise
On right: terrain Generated using Perlin noise

Courtesy: MaximumADHD on Youtube